# Sahaj Bhadja

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## Summary

Software engineer with 2+ years building mobile and distributed products across iOS, Android, and web. Experienced in C++, React Native, TypeScript, Python, real-time messaging, notifications, concurrent programming, performance tuning, testing, and production reliability.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*
- Reduced customer-facing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users by profiling request paths and tuning Azure SQL queries, indexes, and secure access patterns.
- Built fault-tolerant asynchronous workflows for 1,000+ jobs using explicit state transitions, retries, and failure isolation, tripling concurrent throughput while preserving reliable production behavior under sustained load.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Delivered responsive React interfaces and Flask REST APIs for file upload, status updates, history, and asynchronous feedback workflows, supporting 150+ users and 10,000+ daily production requests.
- Designed concurrent processing for 1,000+ uploaded files with real-time job-state updates and fault-tolerant execution, reducing feedback turnaround 60% while maintaining stable API latency during peak demand.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*
- Built reproducible simulation and benchmarking pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through structured evaluation, performance instrumentation, and systematic debugging of asynchronous model behavior.
- Developed a spatial-temporal encoder integrating map geometry and agent context through multi-head attention, improving closed-loop stability 15% while validating correctness and latency across repeated controlled simulation scenarios.

**Aasma Technology Solutions** **India**
*Software Developer Engineer – SecureSync* *August 2023 – July 2024*
- Engineered cross-platform services with 20+ automated unit and integration suites using PyTest, Mockito, and Jenkins, validating reliable high-quality production releases across Windows, Linux, and macOS production environments.
- Optimized distributed REST integrations using object-oriented patterns, filtering, pagination, and caching, improving response times 30% for bandwidth-constrained clients while increasing data-pipeline efficiency 68% under sustained production workloads.

**LST Capitals** **India**
*Web Developer Intern* *November 2022 – January 2023*
- Built React Native clients for iOS and Android with Firebase persistence, real-time state synchronization, image storage, push notifications, and secure sessions across a platform handling 50,000+ records.
- Developed Flask REST APIs for bookings, pricing, and payments, delivering sub-200-millisecond responses and reliable secure webhook-driven real-time synchronization that improved confirmation speed 40% across customer-facing mobile experiences.

## Skills

**Mobile & Languages:** iOS, Android, React Native, C++, TypeScript, JavaScript, Python, React, responsive UI, push notifications
**Messaging & Systems:** REST APIs, WebSockets, JSON, networking, asynchronous programming, concurrency, multithreading, real-time synchronization, microservices, failure isolation
**Data & Performance:** Firebase, Firestore, PostgreSQL, MySQL, SQL Server, Azure SQL, indexing, caching, profiling, latency tuning, performance benchmarking
**Quality & Delivery:** Git, CI/CD, Jenkins, PyTest, Mockito, unit testing, integration testing, observability, debugging, incident analysis, Agile, Scrum, AWS, Azure, Docker

## Projects

**FinNexus** | *TypeScript, WebSockets, MongoDB, Plaid, Kubernetes* | Live
- Built a real-time financial platform supporting 50,000+ users and one million transactions, integrating secure bank linking, WebSocket synchronization, advanced filtering, and reliable scalable containerized production deployment.

**CPU-Scheduling Simulator** | *C++, Simulation, Concurrency, React* | Live
- Implemented six scheduling algorithms in C++ with repeatable simulations and performance visualization, evaluating waiting and turnaround behavior across 1,000+ process configurations while reducing debugging time 20%.

**HMIS** | *TypeScript, REST APIs, Appwrite, Twilio, Sentry* | Live
- Delivered responsive appointment workflows with consent-based API access, secure authentication, automated notifications, and Sentry observability, improving reliability and issue visibility across daily global customer-facing healthcare production experiences.